

Scaling Book Exercises: Section 12, Quiz 4

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Quiz: Collectives

Source scan: ../handwritten/section_12_handwritten.pdf, pages 7–10

Reference: <https://jax-ml.github.io/scaling-book/gpus/>

Question 1: AllGather over one SU

Let the SU contain M nodes with N GPUs per node, and let the final gathered array have B bytes.

At one node-level switch:

1. Its N GPUs contribute $N(B/MN) = B/M$ bytes.
2. The switch sends those B/M bytes toward the spine.
3. It receives $B(M - 1)/M$ bytes from the other nodes.
4. It sends $B - B/(MN)$ bytes to each of its N GPUs, for $NB - B/M$ bytes total.

Thus the node switch handles

$$B \text{ bytes ingress,} \quad NB \text{ bytes egress.}$$

Its aggregate egress bandwidth is $N(450 \text{ GB/s})$, so

$$T_{\text{node}} = \frac{NB}{N(450 \times 10^9)} = \frac{B}{450 \times 10^9}.$$

At the spine layer, all M nodes contribute B/M , giving B total ingress. Each node needs the other $M - 1$ shards, giving

$$B \text{ ingress,} \quad B(M - 1) \text{ egress.}$$

With aggregate egress $M(400 \text{ GB/s})$,

$$T_{\text{spine}} = \frac{B(M - 1)}{M(400 \times 10^9)}.$$

Question 2: Single-node SHARP AllReduce

For an in-network AllReduce of B bytes across N GPUs:

1. The ReduceScatter phase ingresses $B(N - 1)$ bytes.
2. The switch returns B reduced bytes in total.

3. The residual contribution adds another B bytes of ingress.
4. Multicast in the AllGather phase egresses $B(N - 1)$ bytes.

Therefore

$$\boxed{BN \text{ bytes ingress and } BN \text{ bytes egress.}}$$

Dividing by aggregate switch bandwidth NW_{link} gives

$$\boxed{T_{\text{AR}} = \frac{B}{W_{\text{link}}}.}$$

Question 3: SHARP AllReduce with two sharding axes

Consider

$$\text{AllReduce}_X(\text{bf16}[D_X, F_Y])$$

within one N -GPU node. Let $B = 2DF$ denote the global array size. Sharding the non-reduced dimension F by Y reduces the switch traffic by Y , giving NB/Y bytes of ingress and the same amount of egress. Hence

$$\boxed{T_{\text{comms}} = \frac{NB/Y}{NW_{\text{link}}} = \frac{2DF}{Y W_{\text{link}}}.}$$

If the collective spans more than one node, the node-level switch must send the full B bytes upward, rather than B/Y , because the Y shards must remain distinct. The upper network level can therefore become the bottleneck.

Question 4: Spine-level AllReduce

For $Y = 256$, the sharding spans a complete SU and can use the 12.8 TB/s of spine bandwidth available per SU. Assuming SHARP in-network reduction,

$$\boxed{T_{\text{AR}} = \frac{2DF}{12.8 \times 10^{12}} \text{ seconds.}}$$

Question 5: Exact two-node AllGather

The exact node-level time for eight GPUs is

$$T_{\text{node}} = \frac{B(8 - 1)}{8(450 \times 10^9)} = \boxed{\frac{B}{514 \times 10^9}}.$$

Across two nodes,

$$T_{\text{cross}} = \frac{B(2 - 1)}{2(400 \times 10^9)} = \boxed{\frac{B}{800 \times 10^9}}.$$

Because $B/(514 \times 10^9) > B/(800 \times 10^9)$, the intra-node phase is the bottleneck in this two-node case.